

SCH4U SYLLABUS

Course Title : Chemistry
Grade : 12
Credit Value : 1.0
Instructor : Istikbal Turut
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Phone : 416 838 90 66
Office : Room 203
Instructor Office Hours : 15:25-16:00

Number of Scheduled Hours : 110 hours
Number of Periods : 10 periods a week, 19 weeks

OUTLINE OF COURSE CONTENT

Unit	Titles and Descriptions	Time and Sequence
Unit 1	Organic Chemistry The unit begins with a discussion of the nature of organic chemistry and moves on to examine the topics: hydrocarbons, alcohols and ethers, amines, carbonyl compounds, organic polymers and living organisms.	21 hours
Unit 2	Structure and Properties Early research on atoms begins this unit which then pursues electron configuration and the properties of different chemical bonds.	19 hours
Unit 3	Energy Changes and Rates of Reactions Several topics are studied in this unit including: Physical and Chemical Reactions, Collision Theory, Rate Reaction, Potential Energy, Reaction Mechanisms and Rate law, Hess's Law and	18 hours

	Nuclear Reactions.	
Unit 4	Chemical Systems and Equilibrium Entropy and Acids and Bases are the main pursuits of this unit.	30 hours
Unit 5	Electrochemistry Oxidation states, oxidizing and reducing agents and galvanic cells form the bulk of the topics covered by the Electrochemistry Unit.	19 hours
	Final Evaluation	3 hours
	TOTAL	110 hours

TEACHING AND LEARNING STRATEGIES

Along with some of the strategies noted in the assessment for, as and of learning charts below, strategies will include (but not limited to):

- demonstrations
- structured discussions
- real world problem solving
- socratic teaching
- video support
- student-specific tutorials and one-on-one help
- peer teaching and support
- math talk
- critical pedagogies
- smartboard

Assessment Methods and Tools:

<i>Term Assessment and Evaluation: 70%</i> (Tests, Exams, Quizzes, Lab Reports, Lab Exams)		
Category	Description	Weighting
Knowledge and Understanding	Knowledge of content (e.g., facts, terms, definitions and procedures.) Understanding of content (e.g., concepts, principles, theories, relationships and methodologies)	30%
Thinking and Inquiry	Planning skills (e.g., focusing research, gathering information, selecting strategies,	20%

	organizing a project) Processing skills (e.g., analyzing, interpreting, assessing, reasoning, gathering ideas, evaluating, seeking a variety of perspectives, forming conclusions)	
Communication	Expression of original ideas and information (e.g., logical organization) in oral, visual, and written forms	20%
Application/Making connection	The use of the knowledge and skills to make connections within and between various contexts.	30%
Final Exam: 30%		

The Final Grade:

The evaluation of the student's achievement in this course is based on the student's achievement of the curriculum expectations. The percentage grade represents the student's overall achievement and reflects the corresponding level of achievement as described in the Achievement chart for this discipline. A credit will be granted if the final percentage awarded is 50% or more. The final grade will be determined as follows:

- 70% of the grade will be based upon evaluations conducted throughout the course. This portion of the grade will reflect the student's most consistent level of achievement throughout the course, although special consideration will be given to more recent evidence of achievement.
- 30% of the grade will be based on a final exam administered at the end of the course. The exam will contain a summary of information from the course and the student's reports.

Distribution of points used for course evaluation

Exercise	# assigned	Points each	Total points
Homework/Classroom	5	2	10
Quizzes/Assignments	5	2	10
Tests	5	10	50
Final Exam	1	30	30